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“Love Over Fear”: How Nostalgic Songs and Relationship Types Buffer Death Anxiety

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ABSTRACT

The fear of death is a fundamental human anxiety. From the perspective of terror management theory (TMT), this study examines the potential of music to alleviate death anxiety. We conducted a 2 (salience: mortality vs. control) $\times$ 2 (song type: nostalgic vs. non-nostalgic) $\times$ 2 (relationship type: romantic vs. parent-child) between-subjects factorial experiment with 335 middle-aged participants (ages 35–40) who were married, had children, and had living parents. Multiple interaction effects among mortality salience, nostalgia, and relationship type indicated that romantic songs – especially nostalgic romantic songs – were more effective in reducing death anxiety than parent-child relationship songs. The findings highlight the defensive function of romantic relationship songs within the TMT framework, offering new insights for clinical interventions and public mental health practices aimed at managing death anxiety.

Introduction

Death is an inevitable and enduring human concern. As Yalom (2008) states, human beings are “forever shadowed by the knowledge that we will grow, blossom, and inevitably, diminish and die” (p. 1). The fear of death is deeply ingrained in the human psyche, with Jung (1919) suggesting that it resides within people’s collective unconscious. Unlike other species, humans possess an awareness of their mortality (Becker, 1973), which gives rise to death anxiety – a psychological state of unease, distress, and fear triggered by thoughts of one’s eventual death (Lehto & Stein, 2009). Rooted in the innate drive for survival, death anxiety is universal and has been linked to various mental health disorders, including generalized anxiety disorder, depression, and post-traumatic stress disorder (Iverach et al., 2014). In a word, higher levels of death anxiety correlate with poorer overall mental health (Jazaieri et al., 2022; Y. Jin et al., 2023). Given its widespread impact, it is crucial to develop effective interventions to mitigate death anxiety and enhance psychological well-being.

Terror management theory (TMT) (Greenberg et al., 1986) offers a framework for understanding both the origins of and responses to death anxiety. According to TMT, mortality salience, such as encountering a funeral or witnessing an accident, triggers existential anxiety. To cope, individuals reinforce cultural worldviews that provide symbolic meaning and self-worth. For example, religious believers may strengthen their faith to reaffirm life’s significance. In communication studies, researchers have explored how meaningful media content serves as a coping mechanism for mortality salience. Engaging with such content enhances self-worth and reinforces a sense of purpose (Bonus et al., 2022; Goldenberg et al., 1999; Hofer, 2013; Rieger et al., 2015). For instance, Rieger et al. (2015) found that participants exposed to mortality salience showed greater appreciation for meaningful films,

which in turn effectively buffered against death-related anxiety. In an era of widespread digital media access, leveraging media as an intervention for death anxiety presents a practical and efficient approach. This study aims to explore the types of media content that can effectively alleviate death anxiety, contributing to both theoretical insights and real-world applications in mental health.

Notably, among various forms of media, music stands out as a particularly powerful and ubiquitous medium that can offer both entertainment and emotional support. It is widely recognized that music evokes strong emotions, from joy to sorrow (Lonsdale & North, 2011; Schäfer et al., 2013). Beyond these responses, music can also elicit nostalgic feelings strongly (Barrett et al., 2010; Sedikides et al., 2022), often evoking memories of meaningful experiences and providing emotional comfort (Saarikallio, 2011). Janata et al. (2007) also identified that nostalgia is one of the most common emotional responses to music. Importantly, nostalgia serves as a psychological resource with existential defense functions. By recalling cherished past experiences, nostalgia helps individuals connect with enduring cultural traditions and social identities, fulfilling their psychological need for security in the present (Routledge et al., 2011, 2014; Yin et al., 2023). This seems to align with TMT’s claim that cultural worldviews buffer against death anxiety, as their enduring nature offers a sense of permanence that transcends the limits of individual lifespan. As a result, nostalgic music holds strong potential as a medium for alleviating death anxiety.

In addition, since nostalgic experiences often center on significant relationships (Madoglou et al., 2017; Sedikides & Wildschut, 2019; Wildschut et al., 2006), we classify song themes into two categories: romantic relationship and parent-child relationship. This distinction allows us to investigate whether different relationship-oriented nostalgic songs vary

in their effectiveness in buffering against death anxiety. Moreover, according to TMT, individuals facing mortality salience often seek support from close relationships (Cox et al., 2008; Pyszczynski et al., 2015), making relational themes particularly relevant in this context.

Using a 2 (salience: mortality vs. control) $\times$ 2 (song type: nostalgic vs. non-nostalgic) $\times$ 2 (relationship type: romantic vs. parent-child) between-subjects factorial design, this study examines the effects of nostalgic music and relationship-themed songs on mitigating death anxiety, as well as their interactive effects. While previous research has rarely compared how different relationship-oriented songs influence death anxiety, this study provides a nuanced understanding of music's existential buffering mechanisms. By exploring music as a potential intervention for death anxiety, this research offers innovative and practical pathways for reducing distressing thoughts about mortality and enhancing overall well-being.

Literature review

Terror management theory

Terror management theory (TMT) (Greenberg et al., 1986) is an existential perspective that proposes reminders of death, known as mortality salience, trigger anxiety related to human mortality. To counteract this existential distress, individuals suppress death-related thoughts by reinforcing their symbolic self, which provides a sense of meaning and value beyond physical existence. According to the *mortality salience hypothesis*, thoughts of death activate terror management mechanisms, leading individuals to strengthen their cultural worldviews, self-esteem, and close relationships as psychological defenses. The *anxiety buffer hypothesis* further posits that these mechanisms, in turn, mitigate the underlying fear and distress associated with mortality awareness (Pyszczynski et al., 2015). Through this process, individuals invest in enduring symbolic aspects of the self to counteract death anxiety.

Despite extensive research on TMT's cultural coping strategies, surprisingly few studies have directly tested whether mortality salience increases death anxiety – a fundamental assertion of the theory (Abeyta et al., 2014; Routledge & Juhl, 2010). While it is commonly assumed that mortality salience induces death anxiety, empirical evidence on this relationship remains limited. Notably, few studies have directly examined death anxiety as a dependent variable in response to mortality salience, and their findings are mixed. Recent research found that mortality salience could increase death anxiety compared to control condition (S. V. Jin & Ryu, 2022; Wang et al., 2023). However, earlier studies did not find a main effect. For example, Routledge and Juhl (2010) were the first to test whether mortality primes increase death anxiety, finding that mortality salience heightened death anxiety only among individuals with low perceptions of meaning in life. Similarly, Juhl et al. (2010) found no main effect of mortality salience on death anxiety, with increased anxiety observed only among participants with low nostalgia proneness. Likewise, Abeyta et al. (2014) reported that mortality salience heightened death anxiety only for individuals with low self-esteem. These findings

suggest that the relationship between mortality salience and death anxiety may depend on individual differences, rather than being a universal response. Given the limited findings on this issue, this study aims to further investigate the relationship between mortality salience and death anxiety. We propose the following exploratory research question:

RQ1: Does mortality salience increase death anxiety?

The buffering effect of nostalgic songs

According to TMT, when individuals are reminded of their inevitable mortality, they seek comfort in structures that give their existence meaning, preventing death awareness from escalating into death anxiety. Nostalgia, through reflection on and reaffirmation of past experiences, helps individuals reconstruct meaning and strengthen their worldviews, making it a valuable psychological resource for managing existential anxiety. *The New Oxford Dictionary of English* (Pearsall, 1998, p. 1266) defines nostalgia as “a sentimental longing or wistful affection for the past.” Derived from the Greek words *nostos* (return) and *algos* (pain), nostalgia was originally associated with homesickness-induced distress. While once considered a disorder associated with anxiety, sadness, and insomnia, contemporary research recognizes nostalgia as an adaptive psychological resource (Sedikides et al., 2008) and an ambivalent but mostly positive emotion (Leunissen et al., 2021). Nostalgia is a self-referential emotional experience that allows individuals to recall meaningful life events deeply connected to their personal history, such as childhood memories or significant relationships (Routledge et al., 2011). By fostering a sense of continuity between past and present, nostalgia reassures individuals that their lives are meaningful, helping them cope with the uncertainty of the future. Research increasingly supports nostalgia's role in mitigating existential, self, and social threats (Abeyta et al., 2020; Evans et al., 2021; Fleury et al., 2022; Sedikides et al., 2025). First, nostalgia serves an existential function by buffering against survival threats. Reflecting on past experiences strengthens individuals' connection to broader, enduring cultural traditions and social identities, reinforcing the belief that life is filled with meaningful events, free from anxiety associated with future uncertainty (Routledge et al., 2011, 2014; Yin et al., 2023). Experimental studies showed that participants who wrote about nostalgic experiences (compared to general autobiographical or recent positive events) reported greater perceptions of life meaning (Routledge et al., 2012), and induced nostalgia helped lonely people restore meaning in life (Abeyta & Juhl, 2023). Second, nostalgia has a self-oriented function, serving as a repository of positive emotions, enhancing subjective well-being (B. Li et al., 2023; Wildschut et al., 2006). Finally, nostalgia fosters social connectedness, reducing loneliness by strengthening perceived social support (Abeyta et al., 2020; Zhou et al., 2022). By reigniting meaningful relational bonds, it helps people feel reconnected and loved (Wildschut et al., 2006; Zhou et al., 2008). As a result, nostalgia can serve as a buffer against harmful stimuli and have a positive impact on mental health (Wildschut & Sedikides, 2020). There is recent

evidence that nostalgia can alleviate the pain of bereavement (Reid et al., 2021) and boost optimism during the COVID-19 pandemic (Faul & Brigard, 2022). A qualitative study also found that medical staff were able to comfort cancer patients by helping them recall nostalgic memories (Missel et al., 2022).

Despite nostalgia's psychological benefits, its direct effect on reducing death anxiety remains inconclusive. A review of existing studies identified only one that examined death anxiety as a dependent variable, and it found no significant main effect of nostalgia (Juhl et al., 2010). But another relevant study revealed that nostalgia significantly reduced death-thought accessibility (Routledge et al., 2008). However, encouragingly, both studies confirmed an interaction effect between nostalgia and mortality salience. Juhl et al. (2010) found that mortality salience significantly increased death anxiety only among individuals with low nostalgia proneness, while those high in nostalgia showed no significant difference between mortality salience and control conditions. Similarly, Routledge et al. (2008) reported that for participants high in nostalgia proneness (Experiment 2) or those who completed a nostalgia-induction task (Experiment 3), mortality salience did not significantly increase death-thought accessibility. Additional behavioral and neural evidence also supports nostalgia's protective role. Some studies showed that nostalgia could enhance threat processing by amplifying amygdala activation in response to death-related stimuli, potentially preparing individuals to better cope with existential threats (Fleury et al., 2022; Z. Yang et al., 2021). Moreover, confronted with existential threats, individuals demonstrated an increased inclination toward nostalgia, which functioned as a psychological buffer (Barauskaitė et al., 2022). These findings suggest that nostalgia buffers against the negative effects of mortality salience.

Our study examines music as a medium for inducing nostalgia. Prior research strongly supports music as a powerful catalyst for nostalgia, effectively evoking memories of meaningful past experiences (Barrett et al., 2010; Juslin et al., 2008; Sedikides et al., 2022). Thus, we expect that carefully selected nostalgic songs will effectively induce a high level of nostalgic feelings. Moreover, related research suggests that certain music genres can buffer against existential anxiety. For instance, heavy metal music has been shown to mitigate fear of death among its fans (Kneer & Rieger, 2016). Given nostalgia's established psychological benefits, nostalgic music may similarly serve as a defense against death anxiety. Based on prior findings, we propose the following research question and hypothesis:

RQ2: Do nostalgic songs help reduce death anxiety?

H1: Nostalgic songs moderate the effects of mortality salience, such that nostalgia buffers against mortality salience-induced death anxiety.

Different relationship types in songs and their effects on death anxiety

Routledge and his colleagues were pioneers in exploring the interaction between nostalgia and mortality salience, and they

called for future research to consider the differential effects of various types of nostalgic memories (Routledge et al., 2014). Nostalgic narratives are often associated with close others, such as romantic partners, family members, and friends (Madoglou et al., 2017; Sedikides & Wildschut, 2019; Wildschut et al., 2006). Among these, romantic and parent-child relationships are particularly relevant to individuals' experiences with mortality: romantic partners accompany us to the end of life, while family members may either precede us in death, bringing us closer to the reality of mortality, or outlive us, witnessing our eventual passing. This study compares romantic relationship songs and parent-child relationship songs to examine their differential effects on alleviating death anxiety.

TMT posits that when confronted with death threats, individuals seek to enhance meaning in life through self-esteem, cultural worldview, or close relationships (Pyszczynski et al., 2015). Attachment theory (Bowlby, 1973, 1982) similarly emphasizes the anxiety-buffering function of close relationships (Plusnin et al., 2018). For example, Mikulincer et al. (1990) observed that secure attachment effectively reduced death anxiety and fear of death. Weise et al. (2008) further demonstrated that, under mortality salience, participants primed with security attachment exhibited lower aggression in response to terrorism-related issues, indicating that security attachment enhances individuals' ability to regulate death-related fear. The protective role of close relationships has been further supported by recent research during the COVID-19 pandemic (Chen et al., 2021; Mayer & Vanderheiden, 2022; Miller & Ma, 2021). For instance, an experimental study showed that when individuals were exposed to COVID-19 salience, their reliance on close relationships as a defense mechanism increased, and those with higher levels of perceived social support experienced stronger protective effects (Chen et al., 2021). These theoretical and empirical findings all support the buffering effect of close relationships on anxiety. Therefore, through reflecting on and reminiscing about close relationships, individuals can experience feelings of connection and being loved from this relational nostalgia, thereby enhancing their perception of meaning in life. Mikulincer et al. (2003) described romantic partners as "safe havens" that provide support and comfort during periods of distress. Romantic nostalgia refers to reminiscing about past experiences shared with the current romantic partner (Evans et al., 2022). Evans et al. (2022) found that listening to romantic nostalgic songs increased feelings of romantic connectedness, relationship optimism, closeness, satisfaction, compassionate love, and passionate love. Additionally, Swets et al. (2024) found that nostalgia for shared romantic moments helps mitigate relationship conflicts and reinforce commitment. While this study focuses on nostalgia tied to current relationships, memories of former partners may also have positive effects. An experimental study showed that nostalgia about ex-partners evoked more positive than negative emotions, and even improved perceptions of current relationship quality and personal growth (Ai et al., 2023).

While most research has focused on the anxiety-buffering effects of romantic relationships, parent-child relationships also serve a terror management function

(e.g., Cox et al., 2008; Yaakobi et al., 2014). From a TMT perspective, the biological and emotional bonds between parents and children provide existential security, offering a sense of symbolic immortality (Solomon, 2019). Symbolic immortality refers to the belief that an one's existence endures beyond physical death through lasting legacies or representations (Greenberg & Kosloff, 2008). Biological continuity through offspring is one such enduring legacy – we are the continuation of our parents' lives, just as our children will be ours. This sense of connection to future generations serves as a psychological defense against death anxiety. Supporting this, Wisman and Goldenberg (2005) found that when career and reproduction were not in conflict, mortality salience increased participants' desire for offspring. Cox et al. (2008) found that under mortality salience, activating parental-related thoughts reduced death-thought accessibility. Yaakobi et al. (2014) also confirmed that parenthood can have an anxiety-buffering effect. A study based on a Chinese sample further highlighted the terror management function of offspring, especially within a culture that strongly emphasizes lineage continuity (Qi, 2023). In this context, children are viewed not only as carriers of the family line but also as successors expected to fulfill their parents' unrealized aspirations.

However, few studies have directly compared the effects of romantic and parent-child relationships in buffering death anxiety, except for Cox et al. (2008), to our best knowledge. Their findings suggest that attachment style influences individuals' preferences for different attachment figures when confronted with mortality reminders – securely attached individuals rely more on romantic partners, whereas anxiously attached individuals turn to their parents. If we compare the effects of romantic and parent-child songs, we may consider the potential downside of parent-focused songs. Aging parents may themselves be mortality reminders, making nostalgia for them a double-edged sword. For example, Martens et al. (2004) found that participants exposed to images of elderly individuals exhibited higher death-thought accessibility than those shown images of younger people. Mortality salience may also heighten awareness that parents are aging, leading to “upward temporal comparisons” (Bonus, 2021, p. 1096), where individuals contrast their childhood closeness with parents against the present, noticing relational changes. This realization could lead to feelings of loss and transience, reducing nostalgia's potential to buffer against death anxiety. Conversely, romantic partners remain figures who walk alongside us toward life's end, jointly facing existential threats. Thus, romantic relationship songs may be more effective than parent-child in reducing death anxiety. In addition, from an attachment perspective, attachment functions gradually shift from parents to romantic partners in adulthood (Hazan & Zeifman, 1994). Adults usually have stronger attachment bonds with romantic partners than with parents (Dansby Olufowote et al., 2020; Doherty & Feeney, 2004). Based on these insights, we propose the following hypotheses:

H2: Romantic relationship songs are more effective than parent-child relationship songs in alleviating death anxiety.

H3: There is an interaction between song relationship type and mortality salience: (a) Under mortality salience, romantic relationship songs are more effective than parent-child relationship songs in reducing death anxiety; (b) Within the parent-child song condition, mortality salience will lead to higher death anxiety than the control condition.

H4: There is an interaction between nostalgia and song relationship type, such that romantic nostalgic songs are more effective than parent-child nostalgic songs in reducing death anxiety.

H5: There is a three-way interaction among nostalgia, song relationship type, and mortality salience: Romantic nostalgic songs will be more effective than parent-child nostalgic songs in buffering against mortality salience-induced death anxiety.

Pretest

To effectively induce nostalgia, we selected participants from the same age cohort to ensure they shared collective memories of classic songs. The study focused on middle-aged adults aged 35–40, as 35 is commonly recognized as the threshold of midlife (Freund & Ritter, 2009). According to Erikson's work on life span developmental tasks, midlife is marked by the generativity versus stagnation crisis, where death anxiety peaks before declining in later life (Erikson, 1982; Erikson et al., 1986). Empirical research further supports this, showing a negative correlation between age and death anxiety, with middle-aged individuals experiencing more death-related thoughts than older adults (Chopik, 2017). Additionally, concerns about aging, health decline, and physical deterioration become more salient in midlife, exacerbating anxiety about mortality (Lynch, 2000). Given these factors, our study specifically examined death anxiety in this population. To control for variations in participants' social relationships, which might influence responses to different song types, we selected only married individuals with children whose parents were still alive.

To ensure the effectiveness of nostalgic songs, a pilot study was conducted. Bonus (2021) suggested that nostalgic songs should meet two criteria: (1) they should be at least 10 years old, and (2) they should have been released when participants were in early adolescence, a period when music strongly influences autobiographical memory. For the nostalgic condition, we first identified songs from online popular classic song lists. We then refined this selection through purposive sampling, drawing on our knowledge of the Chinese music scene and insights from interviews and consultations with individuals from the target demographic, to ensure we included the most representative classic songs. For the non-nostalgic condition, we selected recent songs unlikely to evoke nostalgia. In total, we initially selected 16 songs—four in each category (nostalgic-romantic, non-nostalgic-romantic, nostalgic-parent-child, and non-nostalgic-parent-child). Some of these songs were featured in China's Spring Festival Gala, a widely watched New Year's Eve program that evokes strong emotional connections. The relationship themes were easily identifiable through the

lyrics, which included phrases like “sweet love” for the romantic songs and “dear mother/father” for the parent-child songs. The lyrics of all selected songs expressed deep love for either a romantic partner or a parent.

A total of 119 participants ($M_{\text{age}} = 37.50$, $SD = 1.35$, from 35 to 40; 32 men, 87 women) evaluated these songs. Each participant listened to four 30-second excerpts from one category in random order, featuring the most recognizable segments near the song’s climax. The analysis of variance (ANOVA) results confirmed that nostalgic songs ($M = 6.00$, $SD = 0.69$) elicited significantly stronger nostalgic feelings than non-nostalgic songs ($M = 5.33$, $SD = 0.87$; $F(1,117) = 21.32$, $p < .001$, $\eta^2 = .15$) (the items for the variable are provided in the Measures section below). This confirmed the effectiveness of nostalgia induction. Based on participants’ accuracy rates in relationship manipulation check questions (i.e., the associated people evoked by these songs), we ultimately selected two songs from each category that best achieved the targeted relationship induction effect, totaling eight songs. These were used as experimental stimuli in the formal experiment, presented in random order (as listed in Table 1). Additionally, to ensure gender inclusivity in romantic and parent-child relationships, the final selection featured diverse narrative perspectives, minimizing potential biases.

Method

Participants

The experiment was conducted in March 2025 via Credamo (<https://www.credamo.com/#/>), one of China’s largest survey platforms. An a priori power analysis using G*Power (effect size $r = .33$, $\alpha = .05$, power = .80) determined a required minimum sample size of 75 participants (Burke et al., 2010).

A total of 335 participants completed the study and passed the manipulation check, including 87 men and 248 women. The age of the sample ranged from 35 to 40 years ($M = 37.39$, $SD = 1.39$). Among them, 89.0% held at least a bachelor’s degree, and 69.3% had a monthly income of at least 10,000 RMB. Each participant was married with children, with both of their parents alive, and received 9.5 RMB for participation.

Procedure and stimuli

This study employed a 2 (salience: mortality vs. control) $\times$ 2 (song type: nostalgic vs. non-nostalgic) $\times$ 2 (relationship type: romantic vs. parent-child) between-subjects factorial design. Participants were randomly assigned to one of eight conditions. The experiment lasted approximately 10 minutes.

Mortality salience was induced through two open-ended questions: (1) “Describe in detail the emotions that arise when you think about your own death.” (2) “Write about the physical changes you anticipate experiencing during the dying process” (Rosenblatt et al., 1989). The control group answered similar questions about dental pain.

A delay task followed the mortality salience induction. Numerous studies on TMT have demonstrated that individuals initially suppress death-related thoughts and deny death after mortality salience induction. However, after completing a delay task, individuals exhibit heightened accessibility of death-thoughts, and engage in defensive mechanisms linked to meaning-making and cultural worldviews (Abeyta et al., 2014; DeWall & Baumeister, 2007; Routledge et al., 2008). Furthermore, anxiety, an inherently uncertain emotion, is also more likely to emerge after a delay task (Abeyta et al., 2014). Therefore, in this study, after mortality salience induction, participants watched a landscape video and answered demographic and individual predisposition measures (e.g., attachment level), which are unrelated to the dependent variable. The delay task lasted approximately three minutes.

Participants then completed the song-listening task. Each participant listened to two 30-second song excerpts in a random order. After listening to each song, they completed a measurement of variables related to the song.

Randomization check

Chi-square test and one-way ANOVA confirmed that participants were randomly assigned across conditions. The results showed no significant differences among participants assigned to the eight conditions with respect to sex ($\chi^2(7) = 1.55$, $p > .05$) or age ($F(7, 327) = 0.89$, $p > .05$).

Measures

Death anxiety

We used the Your Own Death subscale from the Collett-Lester Fear of Death Scale (Lester & Abdel-Khalek, 2003). Participants rated their anxiety about seven items (e.g., “The total isolation of death,” “The shortness of life”) on a 7-point Likert scale (1 = *not anxious at all*, 7 = *extremely anxious*; $M = 4.06$, $SD = 1.49$, Cronbach’s $\alpha = .93$).

Nostalgic feelings

We measured nostalgia using three items from Wildschut et al. (2006): “Right now, I am feeling quite nostalgic,” “Right now, I am having nostalgic thoughts,” and “I feel nostalgic at the moment.” Responses were rated on a 7-point scale (1 =

Table 1. Song stimulus in each condition.

Nostalgia	Relationship	Chinese Song	Release Year
Yes	Romantic	Intimate lover/知心爱人	1997
		Love you for ten thousand years/爱你一万年	1999
	Parent-child	Go home often/常回家看看	1999
No		Lupinus/鲁冰花	1991
	Romantic	I’m calling you from New York/我在纽约打电话给你	2023
		Plum sauce/乌梅子酱	2022
	Parent-child	Father and son/父子	2016
		A family of love/一家人相亲相爱	2024

strongly disagree, 7 = *strongly agree*; $M = 5.87$, $SD = 0.74$, Cronbach's $\alpha = .80$).

To control for potential differences between experimental stimuli, we included the following covariates:

Familiarity

Adapted from Chou and Lien (2014): "This song sounds familiar" (1 = *strongly disagree*, 7 = *strongly agree*; $M = 5.96$, $SD = 1.16$).

Likeability

Adapted from Chou and Lien (2014): "I like this song" (1 = *strongly disagree*, 7 = *strongly agree*; $M = 6.08$, $SD = 0.84$).

Happiness

Measured using a single semantic differential item: "This song is sad/happy" (1 = *sad*, 7 = *happy*; $M = 4.13$, $SD = 1.03$). Their responses were averaged across the two songs for all the above variables (Bonus, 2021).

Attachment level

We also included participants' attachment to their partners/parents as a covariate. Those in the romantic song condition rated their attachment to partners, while those in the parent-child song condition rated their attachment to parents. Six items from the WHOTO scale were used, such as "My partner is the person I most like to spend time with," and "My partner is the person I would count on for advice" (Fraley & Davis, 1997) (1 = *strongly disagree*, 7 = *strongly agree*; $M = 6.08$, $SD = 0.61$, Cronbach's $\alpha = .73$).

Manipulation check

The nostalgia manipulation was successful. The ANOVA results confirmed that nostalgic songs ($M = 5.97$, $SD = 0.71$) induced significantly stronger nostalgic feelings than non-nostalgic songs ($M = 5.77$, $SD = 0.76$; $F(1,333) = 6.20$, $p < .05$, $\eta^2 = .02$). To ensure the effectiveness of relationship-type manipulation, participants were asked to report whom they thought of while listening to the songs. In the romantic song condition, only those who recalled their partner were retained. In the parent-child condition, participants who recalled either their parents or children were included. Among those in the parent-child song condition, 69.5% thought of their parents, while 29.9% recalled both parents and children.

Results

Due to the sex skew in our sample, we first tested for the effect of sex. One-way ANOVAs confirmed that no significant differences between male and female participants in nostalgic feeling ($F(1,333) = 0.85$, $p = .358$) and death anxiety ($F(1,333) = 1.92$, $p = .167$).

To control for potential confounding variables, we included familiarity, likability, happiness and attachment level as covariates in a three-way ANCOVA conducted in R 4.2.2. The results did not support the main effects of mortality salience ($F(1,323) = 0.00$, $p = .99$), nostalgia ($F(1,323) = 0.09$, $p = .76$), or song relationship type ($F(1,323) = 0.91$, $p = .34$) on death anxiety. Thus,

mortality salience did not significantly increase death anxiety (RQ1), nostalgia did not significantly alleviate it (RQ2), and the type of song relationship did not produce a significant difference in reducing death anxiety, leading to the rejection of H2. Additionally, the interaction between nostalgic songs and mortality salience was not significant ($F(1,323) = 0.51$, $p = .47$), resulting in the rejection of H1.

However, we found a significant interaction between mortality salience and song relationship type ($F(1,323) = 9.12$, $p < .01$, $\eta^2 = .03$), as illustrated in Figure 1. Post-hoc LSD tests revealed that romantic relationship songs ($M = 3.77$, $SD = 1.58$) significantly reduced death anxiety induced by mortality salience compared to parent-child relationship songs ($M = 4.34$, $SD = 1.54$; $t(323) = 2.46$, $p < .05$). Additionally, mortality salience ($M = 4.34$, $SD = 1.54$) significantly increased death anxiety in participants who listened to parent-child songs compared to the control group ($M = 3.89$, $SD = 1.46$; $t(323) = -2.07$, $p < .05$), supporting both H3a and H3b.

Furthermore, a significant interaction was observed between song relationship type and nostalgia ($F(1,323) = 9.55$, $p < .01$, $\eta^2 = .03$), as shown in Figure 2. Post-hoc LSD tests indicated that nostalgic romantic songs ($M = 3.81$, $SD = 1.49$) were more effective in reducing death anxiety than nostalgic parent-child songs ($M = 4.35$, $SD = 1.45$; $t(323) = 2.64$, $p < .01$), supporting H4.

Finally, the three-way interaction between nostalgia, song relationship type, and mortality salience was also significant ($F(1,323) = 5.06$, $p < .05$, $\eta^2 = .01$), as depicted in Figure 3. Post-hoc LSD tests revealed that nostalgic romantic songs ($M = 3.84$, $SD = 1.47$) were more effective in mitigating death anxiety induced by mortality salience than nostalgic parent-child songs ($M = 4.41$, $SD = 1.51$; $t(323) = 2.22$, $p < .05$), supporting H5.

Discussion

Key findings

This study examined the effects of mortality salience, nostalgic song, and song relationship type on death anxiety, as well as their interactions. While none of the three factors had

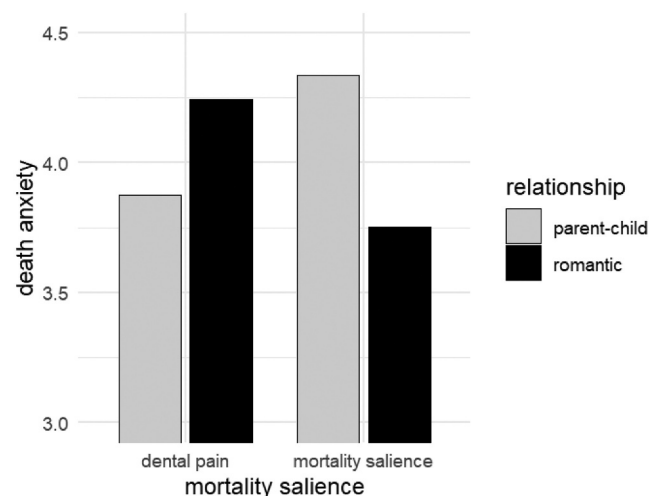


Figure 1. Interaction effect of mortality salience and relationship type on death anxiety.

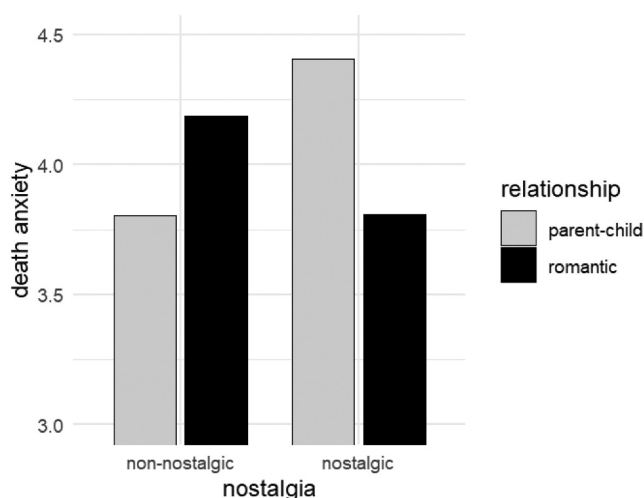


Figure 2. Interaction effect of nostalgia and relationship type on death anxiety.

a significant main effect, we found notable interactions between song relationship type and the other two experimental variables. Specifically, romantic relationship songs significantly buffered the increase in death anxiety caused by mortality salience, whereas mortality salience significantly heightened death anxiety among participants who listened to parent-child songs. Additionally, nostalgic romantic songs were more effective in reducing death anxiety than nostalgic parent-child songs. Furthermore, a significant three-way interaction further revealed that nostalgic romantic songs provided stronger buffering effects against mortality-induced death anxiety than nostalgic parent-child songs. Overall, romantic relationship songs proved more effective in mitigating death anxiety.

Our findings did not support TMT's critical assertion that mortality salience directly triggers death anxiety, aligning with previous studies (Abeyta et al., 2014; Juhl et al., 2010; Routledge & Juhl, 2010). However, this does not necessarily undermine TMT's core premise. Research suggests that individuals may unconsciously regulate death-related distress through positive affective tuning, a coping strategy that shifts attention toward positive emotions to prevent overwhelming

fear (DeWall & Baumeister, 2007). In DeWall and Baumeister's (2007) study, participants exposed to mortality salience used more positive emotion-related words in a word-stem completion task and were more likely to associate words with positive emotions in a word similarity judgment task. This suggests that while death anxiety may be initially activated, individuals may automatically engage in defense mechanisms that suppress it, making it difficult to detect through direct measurement. Another possible explanation is that humans have a natural tendency to avoid thoughts of death, leading them to downplay or deny their actual level of death anxiety (Abeyta et al., 2014). Future research should adopt more subtle, indirect measures that are less likely to be consciously recognized by participants to assess death anxiety. Further investigation is needed to determine whether mortality salience reliably induces death anxiety, as TMT suggests, and under what conditions this effect may be observed.

While our study did not find significant main effects of mortality salience, nostalgia, or song relationship type on death anxiety, we identified key interaction effects that suggest a set of media content strategies with the potential to mitigate death anxiety. First, we found a significant interaction between mortality salience and song relationship type. Under mortality salience, romantic relationship songs significantly reduced death anxiety compared to parent-child songs. Although both parental attachment and the role of parenthood can both serve as psychological refuges in the face of mortality (Cox et al., 2008; Yaakobi et al., 2014), romantic relationships appear to be more effective. One reason is that, in adulthood, a person's primary attachment figure typically shifts from parents to their romantic partner, forming the most intimate and secure connection (Dansby Olufowote et al., 2020; Doherty & Feeney, 2004; Hazan & Zeifman, 1994). This is particularly true for married individuals, as spouses share life's trajectory and face mortality together. In contrast, parent-child relationships are often accompanied by feelings of regret – either the inability to care for aging parents or the sorrow of not witnessing a child's future. This explains the second aspect of the interaction: among those who listened to parent-child songs, mortality salience significantly heightened death anxiety compared to the control group, whereas this effect was absent in the romantic song

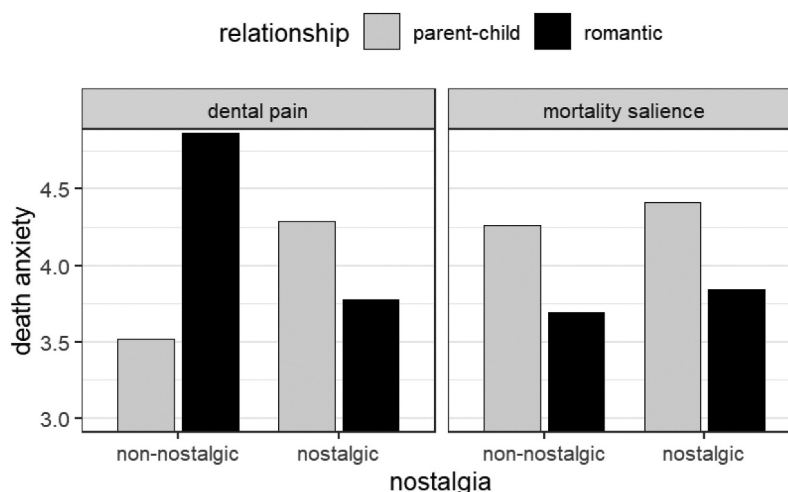


Figure 3. Interaction effect of nostalgia, relationship type and mortality salience on death anxiety.

condition. The thought of losing parents or leaving children behind can evoke a deep sense of sorrow, intensifying death anxiety.

Second, we observed a significant interaction between nostalgia and song relationship type. Nostalgic romantic songs were more effective in reducing death anxiety than nostalgic parent-child songs. Nostalgia often evokes memories of childhood moments with parents or youthful experiences with a romantic partner. However, when people compare their childhood memories of parents to their parents' current aging state, they may experience upward temporal comparisons (Bonus, 2021), noticing a decline in their parents' health and perceiving death as increasingly imminent. This realization can heighten death anxiety. In contrast, listening to romantic relationship songs reinforces emotional bonds, reigniting past romantic passion and strengthening current relationships (Evans et al., 2022; Fetterman & Evans, 2023). According to TMT, reinforced social bonds buffer against death anxiety, explaining the stronger protective effect of nostalgic romantic songs.

Finally, we found a significant three-way interaction among nostalgia, song relationship type, and mortality salience. Specifically, nostalgic romantic songs were more effective than nostalgic parent-child songs in mitigating the increase in death anxiety caused by mortality salience. This suggests that death anxiety regulation depends on both emotional connection and time perception. From an emotional perspective, nostalgic parent-child memories may activate loss-oriented cognition (Stroebe & Schut, 1999), leading individuals to compare past images of their parents with their current aging state, reinforcing the inevitability of death. In contrast, nostalgic romantic memories are more likely to trigger a restoration-oriented mechanism (Stroebe & Schut, 1999), deepening emotional bonds with a romantic partner and positioning intimacy as a psychological resource for coping with existential threats. From a temporal perspective, reconstructing past romantic memories enhances self-continuity (Sedikides et al., 2015), which helps individuals resist the existential fragmentation induced by mortality salience. In contrast, nostalgia parent-child memories may evoke an existential contrast rooted in the intergenerational temporal gap. Parents, whose lives predate the individual, can serve as reminders of aging and one's own eventual decline (Bowlby, 1980).

However, we did not find a significant interaction between mortality salience and nostalgic songs, meaning our study did not confirm the buffering effect of nostalgia reported in previous research (Juhl et al., 2010; Routledge et al., 2008). A possible explanation lies in differences in nostalgia induction methods. Previous studies either treated nostalgia proneness as an experimental factor or used nostalgia writing tasks, while our study relied on music to evoke nostalgia. The differences in how nostalgia was induced may have influenced the effect. Future research should explore how various nostalgia induction methods shape the relationship between nostalgia and death anxiety.

Strengths and limitations of the study

To our best knowledge, this study is the first to examine the interaction between mortality salience, nostalgic songs, and

song relationship type. Building on prior research on nostalgia's buffering effects against mortality salience (Juhl et al., 2010; Routledge et al., 2008), we introduced music as a medium for nostalgia induction, expanding the conceptualization of nostalgic experiences. Notably, no previous studies have compared the differential effects of romantic versus parent-child nostalgia, and few have explored the interaction effects between relational types and mortality salience. Our findings highlight significant variations across relationship types, underscoring the unique role of romantic relationships in alleviating death anxiety. These insights suggest that interventions targeting death anxiety should pay greater attention to the protective function of romantic bonds.

From a media content strategy perspective, our results offer valuable implications for clinical interventions and psychological well-being practices related to death anxiety. First, in mortality salience contexts, careful selection of music is crucial to avoid exacerbating anxiety. For instance, in hospice care and psychological support for critically ill patients, parent-child-themed music should be avoided in favor of romantic relationship songs. Second, community-based music therapy programs could leverage romantic or nostalgic romantic songs to enhance individuals' resilience and perceived life expectancy. We also recommend that couples create a "romantic nostalgia playlist," incorporating songs they listened to during their courtship – "our songs" – to strengthen their emotional bonds and collectively buffer existential anxiety, including grief from bereavement. Lastly, in times of public crisis (e.g., pandemics, or natural disasters), media platforms could prioritize the dissemination of nostalgic romantic music to mitigate collective anxiety. Platforms could also integrate music-based interventions by recommending nostalgic romantic songs when users search for death-related keywords, helping foster a more supportive digital environment. This approach may potentially mitigate suicidal ideation in individuals with psychiatric disorders. As for a future direction, we recommend exploring the potential benefits of passive nostalgic music interventions (e.g., listening to a song together) versus active nostalgic music interventions (e.g., singing a song together), building on music therapy research that highlights the distinct effects of active and passive rhythmic interventions (McPherson et al., 2019).

Despite its contributions, this study has several limitations that warrant further exploration. First, our sample was recruited through a well-known and professional survey platform in China (Y. Li et al., 2025; Tang et al., 2023; X. Yang et al., 2024), but women far outnumbered men, which may affect the generalizability of the findings. This may be because female users are more likely to be attracted to intimacy and feeling-related survey tasks. However, this sampling skew may be acceptable considering that sex did not significantly affect death anxiety and nostalgic feelings in this study. Future research should explore different demographic groups, such as older adults or terminally ill patients, and individuals whose parents are no longer alive, as they may respond differently. Cultural context also matters because perceptions of death vary widely across ethnicities, religions, and countries, highlighting the need to include more diverse participant groups. Second, future studies could broaden the scope of nostalgia- or

relationship-based experimental stimuli. While our study focused on music, other media formats such as films, photographs, and short videos could provide alternative avenues for investigation. Different methods of inducing nostalgia should also be considered. For example, researchers in psychology have employed a more tailored approach by asking participants to select songs they found most nostalgic and then listen to them (Cho et al., 2023). Finally, as Newman and his colleagues have cautioned, the external validity of laboratory-induced nostalgia effects should be carefully considered (Newman & Sachs, 2020; Newman et al., 2020). Their research, using daily diary and ecological momentary assessment techniques, suggests that nostalgia in everyday life may lead to more negative outcomes than previously assumed. A key reason for the discrepancy between experimental and real-world findings is that laboratory-induced nostalgia tasks often prompt participants to recall their most nostalgic experiences, which are inherently filtered through an optimism bias, thereby enhancing well-being. However, our study controlled for the happiness level of the selected songs, which may have mitigated this bias to some extent.

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